

1. Let us consider the following surface:

$$\begin{aligned}x(u, v) &= u - \frac{u^3}{3} + uv^2 \\y(u, v) &= v - \frac{v^3}{3} + vu^2 \\z(u, v) &= u^2 - v^2 \\u &\in [-25, 25], \quad v \in [-25, 25]\end{aligned}$$

Draw the surface! Draw the point corresponding to $u = 10$ and $v = 15$ along with the corresponding isocurves. Draw the normal vector of the surface at P .

```
% Q1
syms u v
x(u,v) = u - u^3 / 3 + u * v^2;
y(u,v) = v - v^3 / 3 + v * u^2;
z(u,v) = u^2 - v^2;
fsurf(x,y,z,[-25 25 -25 25]);

u0 = 10;
v0 = 15;
hold on
plot3(x(u0,v0),y(u0,v0),z(u0,v0),'r*')

sx(u) = x(u,v0);
sy(u) = y(u,v0);
sz(u) = z(u,v0);
fplot3(sx,sy,sz,[-25 25])

cx(v) = x(u0,v);
cy(v) = y(u0,v);
cz(v) = z(u0,v);
fplot3(cx,cy,cz,[-25 25])

sxd(u) = diff(sx,u);
syd(u) = diff(sy,u);
szd(u) = diff(sz,u);
vec = [sxd(u0),syd(u0),szd(u0)];

cxd(v) = diff(cx,v);
cyd(v) = diff(cy,v);
czd(v) = diff(cz,v);
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```
vec2 = [cxd(v0),cyd(v0),czd(v0)];
normv = cross(vec,vec2);
quiver3(x(u0,v0),y(u0,v0),z(u0,v0),normv(1),normv(2),normv(3));
```

2. Let us consider the surface

$$z = \sqrt{1 - x^2 - 0.5y^2}.$$

Draw the surface! Draw the point on the surface corresponding to $x = 0.5$ and $y = 0.2$.

```
% Q2
syms x y
F(x,y) = sqrt(1 - x^2 - 0.5 * y^2);
fsurf(F)

x0 = 0.5; y0 = 0.2;
hold on
plot3(x0,y0,F(x0,y0),'r*')
```

3. Let us consider the surface

$$\sin(x) + \frac{\cos(y)}{x} - z = 0.$$

Draw the surface! Draw the intersection of the surface with the xy plane.

```
% Q3
syms x y z
F(x,y,z) = sin(x) - cos(y) / x - z;
fimplicit3(F)

hold on
fimplicit(F(x,y,0),'LineWidth',10)
```

4. Let us define the following 3 planes:

$$x + y - z = 0, \quad x - 2y + 3z = 4, \quad 2x - 0.5y + 4z = -2.$$

Draw them with different colors!

```
% Q4
syms x y z
F(x,y,z) = x + y - z;
G(x,y,z) = x - 2*y + 3*z - 4;
H(x,y,z) = 2*x - 0.5*y + 4*z + 2;
```

```
fimplicit3(F,'r');
hold on
fimplicit3(G,'g');
Fimplicit3(H,'b');
```

5. Let us have curves

$$\begin{aligned} p(u) &= (1 - u) P_1 + u P_2 \\ r(u) &= (1 - u) R_1 + u R_2 \\ u &\in [0, 1] \end{aligned}$$

where $P_1 = (0, 0, 0)$, $P_2 = (0, 1, 1)$, and $R_1 = (1, 0, 1)$, $R_2 = (1, 1, 0)$. Let us define the parametric surface as follows:

$$\begin{aligned} s(u, v) &= (1 - v) p(u) + v r(u) \\ u &\in [0, 1], \quad v \in [0, 1] \end{aligned}$$

Draw the given two curves and the given surface on the same figure.

```
% Q5
syms u
px = [0, 0, 1, 1];
py = [0, 1, 0, 1];
pz = [0, 1, 1, 0];

plot3(px,py,pz,'r*')
hold on
```

```

ppx(u) = (1-u)*px(1)+u*px(2);
ppy(u) = (1-u)*py(1)+u*py(2);
ppz(u) = (1-u)*pz(1)+u*pz(2);
fplot3(ppx,ppy,ppz,[0 1])

```

```

rx(u) = (1-u)*px(3)+u*px(4);
ry(u) = (1-u)*py(3)+u*py(4);
rz(u) = (1-u)*pz(3)+u*pz(4);
fplot3(rx,ry,rz,[0 1])

```

```

sx(u,v) = (1-v)*ppx(u)+v * rx(u);
sy(u,v) = (1-v)*ppy(u)+v * ry(u);
sz(u,v) = (1-v)*ppz(u)+v * rz(u);
fsurf(sx,sy,sz,[0 1 0 1])

```

6. Draw a polynomial curve of degree 4 that goes through the points (10, 20), (20, 40), (40, 40), (50, 20), (20, 10) when the parameter is 0, 1, 2, 3, and 4 respectively. Draw the tangent vector of the curve when $t = 0.5$

```

% Q6
px = [10, 20, 40, 50, 20];
py = [20, 40, 40, 20, 10];
t0 = 0; t1 = 1; t2 = 2; t3 = 3; t4 = 4;
plot(px,py,'r*')

```

```

syms t a4 a3 a2 a1 a0
cx(t) = a4 * t^4 + a3 * t^3 + a2 * t^2 + a1 * t + a0;
ex = [cx(t0) == px(1), ...
      cx(t1) == px(2), ...
      cx(t2) == px(3), ...
      cx(t3) == px(4), ...
      cx(t4) == px(5)];
mx = solve(ex, [a4 a3 a2 a1 a0]);
x(t) = mx.a4 * t^4 + mx.a3 * t^3 + mx.a2 * t^2 + mx.a1 * t + mx.a0;

```

```

syms b4 b3 b2 b1 b0
cy(t) = b4 * t^4 + b3 * t^3 + b2 * t^2 + b1 * t + b0;
ey = [cy(t0) == py(1), ...
      cy(t1) == py(2), ...
      cy(t2) == py(3), ...
      cy(t3) == py(4), ...
      cy(t4) == py(5)];

```

```
my = solve(ey, [b4 b3 b2 b1 b0]);
y(t) = my.b4 * t^4 + my.b3 * t^3 + my.b2 * t^2 + my.b1 * t + my.b0;
hold on
```

```
fplot(x,y,[t0 t4])
xd(t) = diff(x,t);
yd(t) = diff(y,t);
quiver(x(0.5),y(0.5),xd(0.5),yd(0.5));
```

7. Let us define the points $P_1 = (-2, -2)$, $P_2 = (4, 0)$, $P_3 = (6, -2)$, $P_4 = (10, 2)$. Draw the Hermite arc that goes through these points at the parameter values $-1, 0, 2, 3$ respectively. Draw the tangent vector of the curve when $t = 2$.

```
% Q7
px = [-2, 4, 6, 10];
py = [-2, 0, -2, 2];
t0 = -1; t1 = 0; t2 = 2; t3 = 3;
plot(px,py,'r*')
```

```
syms t a3 a2 a1 a0
cx(t) = a3 * t^3 + a2 * t^2 + a1 * t + a0;
ex = [cx(t0) == px(1), ...
      cx(t1) == px(2), ...
      cx(t2) == px(3), ...
      cx(t3) == px(4)];
mx = solve(ex,[a3 a2 a1 a0]);
```

```
syms b3 b2 b1 b0
cy(t) = b3 * t^3 + b2 * t^2 + b1 * t + b0;
ey = [cy(t0) == py(1), ...
      cy(t1) == py(2), ...
      cy(t2) == py(3), ...
      cy(t3) == py(4)];
my = solve(ey, [b3 b2 b1 b0]);
x(t) = mx.a3 * t^3 + mx.a2 * t^2 + mx.a1 * t + mx.a0;
y(t) = my.b3 * t^3 + my.b2 * t^2 + my.b1 * t + my.b0;
hold on
```

```
fplot(x,y,[t0 t3])
xd(t) = diff(x,t);
yd(t) = diff(y,t);
quiver(x(2),y(2),xd(2),yd(2))
```

8. Let us define the points $P_1 = (-2, -2)$, $P_2 = (6, -2)$, $P_3 = (10, 2)$, and the vector $\mathbf{v} = (6, -4)$. Draw the Hermite arc that goes through these points at the parameter values 0, 1, 1.5, and whose tangent vector at 0 is vector $\mathbf{v}$.

```
% Q8
px = [-2, 6, 10];
py = [-2, -2, 2];
vx = 6;
vy = -4;
t0 = 0; t1 = 1; t2 = 1.5;

syms a3 a2 a1 a0 t b3 b2 b1 b0
cx(t) = a3 * t^3 + a2 * t^2 + a1 * t + a0;
cy(t) = b3 * t^3 + b2 * t^2 + b1 * t + b0;
cxd(t) = diff(cx,t);
cyd(t) = diff(cy,t);

ex = [cx(t0) == px(1), ...
      cxd(t0) == vx, ...
      cx(t1) == px(2), ...
      cx(t2) == px(3)];

ey = [cy(t0) == py(1), ...
      cyd(t0) == vy, ...
      cy(t1) == py(2), ...
      cy(t2) == py(3)];
mx = solve(ex, [a3 a2 a1 a0]);
my = solve(ey, [b3 b2 b1 b0]);
x(t) = mx.a3 * t^3 + mx.a2 * t^2 + mx.a1 * t + mx.a0;
y(t) = my.b3 * t^3 + my.b2 * t^2 + my.b1 * t + my.b0;
plot(px,py,'r*')
hold on

quiver(px(1),py(1),vx,vy)
fplot(x,y,[t0 t2])
```

9. Let us define the points $P_1 = (-2, -2)$, $P_2 = (6, -2)$, and vectors $\mathbf{v}_1 = (6, -4)$ and $\mathbf{v}_2 = (4, 4)$. Draw the Hermite arc that goes through these points at the parameter values 0, 1, and whose tangent vector at 0 is $\mathbf{v}_1$ and at 1 is $\mathbf{v}_2$.

```
% Q9
px = [-2, 6];
py = [-2, -2];
vx = [6, 4];
vy = [-4, 4];
t0 = 0; t1 = 1;
plot(px,py,'r*');
hold on

quiver(px(1),py(1),vx(1),vy(1))
quiver(px(2),py(2),vx(2),vy(2))

syms a3 a2 a1 a0 t
cx(t) = a3 * t^3 + a2 * t^2 + a1 * t + a0;
cxd(t) = diff(cx,t);
ex = [cx(t0) == px(1), ...
      cxd(t0) == vx(1), ...
      cx(t1) == px(2), ...
      cxd(t1) == vx(2)];
mx = solve(ex, [a3 a2 a1 a0]);
x(t) = mx.a3 * t^3 + mx.a2 * t^2 + mx.a1 * t + mx.a0;

syms b3 b2 b1 b0
cy(t) = b3 * t^3 + b2 * t^2 + b1 * t + b0;
cyd(t) = diff(cy,t);
ey = [cy(t0) == py(1), ...
      cyd(t0) == vy(1), ...
      cy(t1) == py(2), ...
      cyd(t1) == vy(2)];
my = solve(ey, [b3 b2 b1 b0]);
y(t) = my.b3 * t^3 + my.b2 * t^2 + my.b1 * t + my.b0;

fplot(x,y,[t0 t1])
```

10. Draw a Bézier curve with the control points (10, 20), (20, 40), (40, 40), (50, 20), (20, 10). Draw its tangent vectors at its beginning and at its end.

```
% Q10
px = [10, 20, 40, 50, 20];
py = [20, 40, 40, 20, 10];
plot(px,py,'r*-')
n = 4;
syms t
cx(t) = 0*t;
cy(t) = 0*t;

for i=0:n
    cx(t) = cx(t) + px(i+1) * nchoosek(n,i) * t^i * (1-t)^(n-i);
    cy(t) = cy(t) + py(i+1) * nchoosek(n,i) * t^i * (1-t)^(n-i);
end
hold on
fplot(cx,cy,[0 1])
cxd(t) = diff(cx,t);
cyd(t) = diff(cy,t);
quiver(cx(0),cy(0),cxd(0),cyd(0))
quiver(cx(1),cy(1),cxd(1),cyd(1))

% C1 continuity hermite arc
px2 = [100, 120];
py2 = [110, 130];
plot(px2,py2, 'r*')

syms a3 a2 a1 a0 t
bx(t) = a3 * t^3 + a2 * t^2 + a1 * t + a0;
bxd(t) = diff(bx,t);
ex = [bx(0) == cx(1), ...
      bxd(0) == cxd(1), ...
      bx(1) == px2(1), ...
      bx(2) == px2(2)];
mx = solve(ex, [a3 a2 a1 a0]);
xx(t) = mx.a3 * t^3 + mx.a2 * t^2 + mx.a1 * t + mx.a0;

syms b3 b2 b1 b0
by(t) = b3 * t^3 + b2 * t^2 + b1 * t + b0;
byd(t) = diff(by,t);
ey = [by(0) == cy(1), ...
      byd(0) == cyd(1), ...
```



```
by(1) == py2(1), ...  
by(2) == py2(2)];  
my = solve(ey, [b3 b2 b1 b0]);  
yy(t) = my.b3 * t^3 + my.b2 * t^2 + my.b1 * t + my.b0;  
fplot(xx,yy,[0 2])
```